JAVA PROGRAMMING PRACTICES

Mobile Number Validation and Exceptions

Basic Information

Week	Week 11
Source file	MobileNumberValidation.java
Concepts	try-catch, throw, throws, built-in and custom exceptions.

Objective / Aim

To implement mobile number validation and exceptions using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Read the mobile number as text.
2. Check the characters and length.
3. Catch the syllabus-specified exceptions and display the result.

UML / Program Structure

```
MobileNumberValidation
LengthNotSufficientException extends Exception
```

Source Code

```
import java.util.Scanner;
class LengthNotSufficientException extends Exception
{
    LengthNotSufficientException(String message)
    {
        super(message);
    }
}
public class MobileNumberValidation
{
    static void validate(String number) throws LengthNotSufficientException
    {
        for (int i = 0; i < number.length(); i++)
        {
            if (!Character.isDigit(number.charAt(i))) throw new NumberFormatException();
        }
        if (number.length() > 10) throw new ArrayIndexOutOfBoundsException();
        if (number.length() < 10) throw new LengthNotSufficientException("Length is less than 10");
    }
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter mobile number: ");
        String number = sc.nextLine().trim();
        try
        {
            validate(number);
            System.out.println("Valid number");
        }
        catch (ArrayIndexOutOfBoundsException e)
        {
            System.out.println("Invalid Mobile Number-ArrayIndexOutOfBounds Exception");
        }
        catch (LengthNotSufficientException e)
        {
            System.out.println("Invalid Mobile Number-LengthNotSufficientException");
        }
        catch (NumberFormatException e)
        {
            System.out.println("Invalid Mobile Number-NumberFormatException");
        }
        sc.close();
```

Source Code (continued)

```
    }
}
```

Compilation and Execution

```
javac MobileNumberValidation.java
java MobileNumberValidation
```

Output Screen

```
Enter mobile number: Valid number
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.